ModelIt!

Teacher's Guide

Why Don't We See Dinosaurs?

What Fossils Tell Us About Long Ago

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-LS4-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 2-LS4-1

Make observations of plants and animals to compare the diversity of life in different habitats.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze fossil evidence to draw conclusions about what extinct animals looked like and why they are no longer alive.
Disciplinary Core Idea	**LS4.A Evidence of Common Ancestry and Diversity** — There are many different kinds of living things in any area, and they exist in different places. Fossils provide evidence about the types of organisms that lived long ago.
Crosscutting Concept	**Stability and Change** — Students explore how changes in habitat can cause animal populations to shrink or go extinct, and how stable habitats support diverse life.

CAST Testing Alignment

MULTIPLE CHOICE:

Scientists found a fossil of a fish in the middle of a desert. What does this tell us about this place long ago?

CONSTRUCTED RESPONSE:

Explain how fossils help scientists learn about animals that lived long ago. Give an example of something a fossil can tell us about an extinct animal.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Long before humans existed, amazing creatures walked, swam, and flew across Earth. We know about them because they left behind fossils — preserved clues in the rocks that tell the story of ancient life.

What Are Fossils?

Fossils are the preserved remains or traces of living things from long ago. When an animal dies and gets buried in mud or sand, sometimes its bones, teeth, or shells get slowly replaced by minerals and turn to stone over millions of years. Trace fossils like footprints, nests, and even poop (called coprolites!) can also be preserved. Each fossil is a clue about ancient life.

Why Did Dinosaurs Go Extinct?

About 66 million years ago, a giant asteroid (a rock from space) hit Earth near what is now Mexico. The impact caused massive fires, dust clouds that blocked the sun, and dramatic climate change. Without sunlight, plants died. Without plants, plant-eating dinosaurs starved. Without plant-eaters, meat-eating dinosaurs starved too. The habitat changed so fast and so much that most dinosaurs could not survive.

Living Relatives

Not all dinosaurs died out! Scientists have discovered that birds are actually living dinosaurs — they evolved from small feathered dinosaurs millions of years ago. When you see a chicken, a robin, or an eagle, you are looking at a modern dinosaur! Fossils helped scientists make this amazing discovery by showing the link between feathered dinosaur fossils and modern birds.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify habitat change as an external component and animal survival and population size as internal components that respond to environmental shifts.

E **Establish Relationships**
Students determine that large habitat changes cause animal survival to decrease (negative), which causes population size to decrease. Small changes allow survival to remain high.

V **Visualize & Model**
Students build a model showing how the size of habitat change affects whether animal populations survive or go extinct.

E **Evaluate Outcomes**
Students run small change, medium change, and huge change scenarios to observe different outcomes for animal populations.

R **Revise & Extend**
Students add food supply or climate speed to explore why some habitat changes are more dangerous than others.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Raise your hand if you have ever wanted to see a real dinosaur. Well, today we are going to figure out what happened to them!"*

Watch For

Show pictures of dinosaur fossils. Pass around a small fossil or replica if available.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to be fossil detectives! We will figure out what happened to the dinosaurs."*

Watch For

Read objectives together. Introduce fossil, extinct, habitat, and evidence with pictures.

Slide 3: The Big Question

3 min

Say: *"Dinosaurs lived on Earth for over 160 million years. That is way longer than humans! So what happened?"*

Watch For

Think-pair-share: What do you think happened to the dinosaurs? Collect theories on the board.

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to understand why some animals survive habitat changes and some do not."*

Watch For

Preview LEVER steps. Explain that this model will help us understand extinction.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What are the parts of our extinction model? Habitat Change is what happens to the environment. The other parts show what happens to the animals."*

Watch For

Guide students to sort Habitat Change as external and Animal Survival and Population Size as internal.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When the habitat changes A LOT, do animals survive better or worse? What happens to the number of animals?"*

Watch For

Draw negative arrow from Habitat Change to Animal Survival. Draw positive arrow from Animal Survival to Population Size.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test! First a tiny change, then a HUGE change — like an asteroid hitting Earth!"*

Watch For

Students predict, then run scenarios. Compare small vs. huge habitat change effects on the population.

Slide 8: What Did We Discover?

7 min

Say: *"Dinosaurs went extinct because the habitat changed too fast and too much. But they left us clues — fossils!"*

Watch For

Show fossil pictures. Compare a T-rex fossil to a chicken skeleton. Fun fact: birds ARE dinosaurs!

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Habitat Change
- **INTERNAL:** Animal Survival, Population Size

Relationships

- **Habitat Change to Animal Survival:** NEGATIVE (-) — When the habitat changes a lot, it becomes harder for animals to find food, water, and shelter. The bigger the change, the harder it is to survive. Small changes are okay, but huge changes can be deadly.
- **Animal Survival to Population Size:** POSITIVE (+) — When animals survive well, the population stays large and healthy. When survival drops, the population gets smaller and smaller. If survival drops to zero, the animal goes extinct — gone forever.

Key Discoveries

- When habitats change a little, most animals can survive and adapt
- When habitats change a LOT (like a huge asteroid hitting Earth), many animals cannot survive and go extinct
- Fossils are our evidence that extinct animals once lived here — they are like nature's history books
- Some animals alive today look similar to fossils from long ago — they are related to ancient animals

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Dinosaurs and humans lived at the same time"	Dinosaurs went extinct about 66 million years before the first humans appeared! They never met each other. Movies and cartoons sometimes show them together, but in real life there is a huge time gap between dinosaurs and humans.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Extinct means the animal was not strong enough"	Extinction is not about being strong or weak. Even the biggest, most powerful dinosaurs went extinct because their habitat changed too fast. Extinction happens when the environment changes so much that the animals cannot find what they need to survive — no matter how strong they are.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Fossils are just regular rocks"	Fossils look like rocks, but they are special — they contain the remains or traces of living things from long ago. The original bone or shell was slowly replaced by minerals over millions of years, turning it to stone. Fossils are rocks with a story inside them!